

Differentiation test · Algebra

Grade 11

40 minutes 100 marks 17 questions + bonus No calculator

7–9	The rules of differentiation	Algebra 11, §1.4 · P1 · 7.2–7.4
10–12	The derivative of a composite function	Algebra 11, §1.5 · P1 · 7.5
17–18	The equations of the tangent and the normal	Algebra 11, §1.6 · P1 · 7.6
3–4 ext	Limits at infinity — extension	Algebra 11, §1.2 · P1 · 7.1 · new on this paper

- Answer on separate paper, and show your working — method carries marks.
- Simplify every derivative. Leave no negative or fractional index in a final answer.

What the paper weighs

TOPIC	MARKS	SHARE
Lessons 3–4, extension · Limits at infinity	12	12%
Lessons 7–9 · The rules of differentiation	46	46%
Lessons 10–12 · The derivative of a composite function	32	32%
Lessons 17–18 · The equations of the tangent and the normal	10	10%

The rules and the chain rule carry 78 of the 100 marks between them, which is what this paper is for. The tangent and the normal, and the limits at infinity, are there so that a derivative is used for something and not only performed. Nothing on the paper needs lessons 1–6, 15–16, 19–22 or 23–25.

A Section A · Four marks each

EASY 40 marks

Differentiate and simplify. One line of working is enough.

1 $y = 4x^3 - 5x + 7$ [4]

ANSWER

$$\frac{dy}{dx} = 12x^2 - 5$$

WORKING

Power rule term by term; the derivative of the constant 7 is zero.

2 $y = \frac{3}{x^2}$ [4]

ANSWER

$$\frac{dy}{dx} = -\frac{6}{x^3}$$

WORKING

Rewrite as $y = 3x^{-2}$, so $\frac{dy}{dx} = -6x^{-3} = -\frac{6}{x^3}$.

The rewrite is the whole question. A student who cannot turn $\frac{3}{x^2}$ into a power cannot start.

3 $y = 5\sqrt{x}$ [4]

ANSWER

$$\frac{dy}{dx} = \frac{5}{2\sqrt{x}}$$

WORKING

Rewrite as $y = 5x^{1/2}$, so $\frac{dy}{dx} = \frac{5}{2}x^{-1/2} = \frac{5}{2\sqrt{x}}$.

4 $y = (2x + 1)(x - 4)$ [4]

ANSWER

$$\frac{dy}{dx} = 4x - 7$$

WORKING

Expand first: $y = 2x^2 - 7x - 4$, so $\frac{dy}{dx} = 4x - 7$. The product rule gives the same: $2(x - 4) + (2x + 1) = 4x - 7$.

5 $y = \frac{x^3 - 2x}{x}$ [4]

ANSWER

$$\frac{dy}{dx} = 2x$$

WORKING

Divide first: $y = x^2 - 2$ for $x \neq 0$, so $\frac{dy}{dx} = 2x$.

The quotient rule works but wastes two minutes. Looking at the expression before reaching for a rule is the point.

6 $y = (3x + 1)^5$ [4]

ANSWER

$$\frac{dy}{dx} = 15(3x + 1)^4$$

WORKING

Chain rule: $5(3x + 1)^4$ times the derivative of the inside, which is 3.

7 $y = (x^2 - 4)^3$ [4]

ANSWER

$$\frac{dy}{dx} = 6x(x^2 - 4)^2$$

WORKING

Chain rule: $3(x^2 - 4)^2$ times the derivative of the inside, which is $2x$.

8 $y = \frac{1}{2x - 5}$ [4]

ANSWER

$$\frac{dy}{dx} = -\frac{2}{(2x - 5)^2}$$

WORKING

Rewrite as $y = (2x - 5)^{-1}$, so $\frac{dy}{dx} = -(2x - 5)^{-2} \cdot 2 = -\frac{2}{(2x - 5)^2}$.

9 $y = \sqrt{4x + 1}$ [4]

ANSWER

$$\frac{dy}{dx} = \frac{2}{\sqrt{4x + 1}}$$

WORKING

Rewrite as $y = (4x + 1)^{1/2}$, so $\frac{dy}{dx} = \frac{1}{2} (4x + 1)^{-1/2} \cdot 4 = \frac{2}{\sqrt{4x + 1}}$.

10 Evaluate: $\lim_{x \rightarrow \infty} \frac{3x^2 + 2x}{x^2 - 5}$ [4]

ANSWER

3

WORKING

Divide top and bottom by x^2 : $\frac{3 + 2/x}{1 - 5/x^2}$. Both $\frac{2}{x}$ and $\frac{5}{x^2}$ tend to 0, leaving $\frac{3}{1} = 3$.

Equal degrees, so the answer is the ratio of the leading coefficients — but the division has to be shown, not quoted.

B Section B · Eight marks each

MEDIUM

40 marks

Method marks are available even when the final answer is wrong.

11 Differentiate $y = \frac{2x - 1}{x^2 + 3}$ and find the gradient of the curve at $x = 1$. [8]

ANSWER

$$\frac{dy}{dx} = \frac{-2x^2 + 2x + 6}{(x^2 + 3)^2}; \text{ gradient} = \frac{3}{8}$$

WORKING

Quotient rule with $u = 2x - 1$, $v = x^2 + 3$, so $u' = 2$ and $v' = 2x$:

$$\frac{dy}{dx} = \frac{2(x^2 + 3) - (2x - 1)(2x)}{(x^2 + 3)^2} = \frac{2x^2 + 6 - 4x^2 + 2x}{(x^2 + 3)^2} = \frac{-2x^2 + 2x + 6}{(x^2 + 3)^2}.$$

$$\text{At } x = 1: \frac{-2 + 2 + 6}{4^2} = \frac{6}{16} = \frac{3}{8}.$$

The expansion $-(2x - 1)(2x) = -4x^2 + 2x$ is where the sign is usually lost.

- 12 $y = x^3 - 3x^2 + 4$. Find the values of x at which the gradient of the curve is 9. [8]

ANSWER

$$x = 3 \text{ and } x = -1$$

WORKING

$$\frac{dy}{dx} = 3x^2 - 6x. \text{ Set it equal to 9: } 3x^2 - 6x = 9, \text{ so } 3x^2 - 6x - 9 = 0 \text{ and } x^2 - 2x - 3 = 0.$$

That factorises as $(x - 3)(x + 1) = 0$, giving $x = 3$ and $x = -1$.

Two answers, not one. Dividing by 3 before factorising saves the arithmetic.

- 13 $y = \sqrt{x^2 + 9}$. Find $\frac{dy}{dx}$ and its value at $x = 4$. [8]

ANSWER

$$\frac{dy}{dx} = \frac{x}{\sqrt{x^2 + 9}}; \text{ at } x = 4 \text{ it is } \frac{4}{5}$$

WORKING

$$\text{Write } y = (x^2 + 9)^{1/2}. \text{ Chain rule: } \frac{dy}{dx} = \frac{1}{2}(x^2 + 9)^{-1/2} \cdot 2x = \frac{x}{\sqrt{x^2 + 9}}.$$

$$\text{At } x = 4: \sqrt{16 + 9} = 5, \text{ so the value is } \frac{4}{5}.$$

The factor $2x$ from the inside function is the mark most often dropped.

- 14 Differentiate $y = \frac{5}{(1 - 2x)^3}$. [8]

ANSWER

$$\frac{dy}{dx} = \frac{30}{(1 - 2x)^4}$$

WORKING

Rewrite as $y = 5(1 - 2x)^{-3}$. Chain rule:

$$\frac{dy}{dx} = 5 \cdot (-3)(1 - 2x)^{-4} \cdot (-2) = 30(1 - 2x)^{-4} = \frac{30}{(1 - 2x)^4}.$$

Two minus signs, and they cancel: the -3 from the power and the -2 from the inside. An answer of $-\frac{30}{(1 - 2x)^4}$ means only one of them was used.

15 Evaluate, showing the division in each case:

(a) $\lim_{x \rightarrow \infty} \frac{2x^2 - 3x + 1}{5x^2 + 4}$ [3]

(b) $\lim_{x \rightarrow \infty} \frac{x^2 + 1}{x^3 - 2}$ [2]

(c) Explain why $\lim_{x \rightarrow \infty} \frac{x^3 + 2x}{4x^2 - 1}$ does not exist. [3]

ANSWER

(a) $\frac{2}{5}$ (b) 0 (c) it increases without bound

WORKING

(a) Divide by x^2 : $\frac{2 - 3/x + 1/x^2}{5 + 4/x^2} \rightarrow \frac{2}{5}$.

(b) Divide by x^3 : $\frac{1/x + 1/x^3}{1 - 2/x^3} \rightarrow \frac{0}{1} = 0$.

(c) Divide by x^2 : $\frac{x + 2/x}{4 - 1/x^2}$. The denominator tends to 4, but the numerator contains x , which grows without bound — so the quotient grows without bound and no finite limit exists.

The three parts are the three degree cases: top = bottom, top < bottom, top > bottom. A student who has only learnt “compare the degrees” can state the answers but cannot earn the method marks.

C Section C • Ten marks each

Full method required.

HARD

20 marks

16 $y = x^2\sqrt{2x - 1}$.

(a) Show that $\frac{dy}{dx} = \frac{x(5x - 2)}{\sqrt{2x - 1}}$. [6]

(b) Hence find the gradient of the curve at $x = 5$. [4]

ANSWER

(a) shown below (b) $\frac{115}{3}$

WORKING

(a) Product rule with $u = x^2$ and $v = (2x - 1)^{1/2}$. By the chain rule $v' = \frac{1}{2}(2x - 1)^{-1/2} \cdot 2 = \frac{1}{\sqrt{2x - 1}}$, so

$$\frac{dy}{dx} = 2x\sqrt{2x - 1} + \frac{x^2}{\sqrt{2x - 1}}.$$

Over the common denominator $\sqrt{2x - 1}$: $\frac{2x(2x - 1) + x^2}{\sqrt{2x - 1}} = \frac{4x^2 - 2x + x^2}{\sqrt{2x - 1}} = \frac{5x^2 - 2x}{\sqrt{2x - 1}} = \frac{x(5x - 2)}{\sqrt{2x - 1}}.$

(b) At $x = 5$: $\sqrt{10 - 1} = 3$ and $x(5x - 2) = 5 \cdot 23 = 115$, so the gradient is $\frac{115}{3}$.

The product rule and the chain rule in one question, and then the algebra that puts the two terms over one denominator — which is where the “show that” is won or lost.

17 The curve $y = \sqrt{2x + 7}$.

(a) Find $\frac{dy}{dx}$. [3]

(b) Find the equation of the tangent at the point where $x = 1$. [4]

(c) Find the equation of the normal at that point, and the coordinates of the point where it crosses the x -axis. [3]

ANSWER

(a) $\frac{dy}{dx} = \frac{1}{\sqrt{2x + 7}}$ (b) $3y = x + 8$ (c) $y = 6 - 3x$, crossing at $(2, 0)$

WORKING

(a) $y = (2x + 7)^{1/2}$, so $\frac{dy}{dx} = \frac{1}{2}(2x + 7)^{-1/2} \cdot 2 = \frac{1}{\sqrt{2x + 7}}$.

(b) At $x = 1$: $y = \sqrt{9} = 3$ and $\frac{dy}{dx} = \frac{1}{3}$. So $y - 3 = \frac{1}{3}(x - 1)$, which tidies to $3y = x + 8$.

(c) The normal gradient is -3 , so $y - 3 = -3(x - 1)$, that is $y = 6 - 3x$. It meets the x -axis where $y = 0$, so $x = 2$ and the point is $(2, 0)$.

The $\cdot 2$ from the inside function in (a) is the mark most often dropped, and it carries through both later parts.

Leaving the tangent as $y = \frac{1}{3}x + \frac{8}{3}$ is not penalised.

Bonus



Attempt this only when the rest is finished. It can make up marks lost elsewhere, but your total is still recorded out of 100.

BONUS

+10 marks

18 $y = \frac{x + 1}{\sqrt{x}}$ for $x > 0$.

(a) Show that $\frac{dy}{dx} = \frac{x - 1}{2x\sqrt{x}}$. [7]

(b) Hence find the point at which the tangent to the curve is horizontal. [3]

ANSWER

(a) shown below (b) $(1, 2)$

WORKING

(a) Split the fraction before differentiating:

$$y = \frac{x}{\sqrt{x}} + \frac{1}{\sqrt{x}} = x^{1/2} + x^{-1/2}.$$

$$\text{Then } \frac{dy}{dx} = \frac{1}{2}x^{-1/2} - \frac{1}{2}x^{-3/2} = \frac{1}{2}x^{-3/2}(x - 1) = \frac{x - 1}{2x\sqrt{x}}, \text{ since } x^{3/2} = x\sqrt{x}.$$

(b) The tangent is horizontal where $\frac{dy}{dx} = 0$, so $x = 1$. Then $y = \frac{1 + 1}{\sqrt{1}} = 2$, and the point is $(1, 2)$.

The quotient rule also works, but it takes twice as long and the simplification at the end is harder. Splitting the fraction first is the skill being tested.